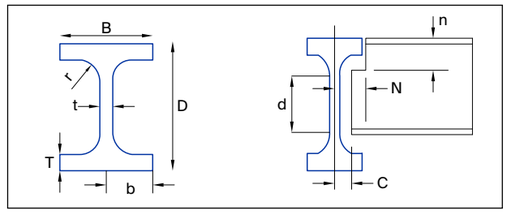
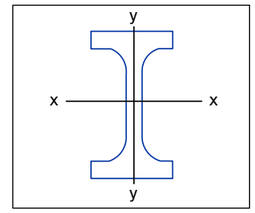
2.4 Metric Beams & Columns (MB & MC)



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
100x50	9.30	100	50	7	5	8	70	11.8	3.57	14.0	5	33	15
100x100	14.8	100	100	7	5	10	66	19.2	7.14	13.2	5	58	17
	16.9	100	100	8	6	8	68	21.6	6.25	11.3	5	57	16
	17.2	100	100	8	6	10	64	21.9	6.25	10.7	5	57	18
125x60	13.2	125	60	8	6	9	91	16.8	3.75	15.2	5	37	17
125x125	23.6	125	125	9	6.5	8	91	30.0	6.94	14.0	5	69	17
	23.8	125	125	9	6.5	10	87	30.3	6.94	13.4	5	69	19
150x75	14.0	150	75	7	5	8	120	17.8	5.36	24.0	5	45	15
150x100	20.7	148	100	9	6	8	114	26.3	5.56	19.0	5	57	17
	21.1	148	100	9	6	11	108	26.8	5.56	18.0	5	57	20
150x150	31.1	150	150	10	7	8	114	39.6	7.50	16.3	6	82	18
	31.5	150	150	10	7	11	108	40.1	7.50	15.4	6	82	21
	37.4	154	151	12	8	11	108	47.7	6.29	13.5	6	82	23
175x90	18.0	175	90	8	5	8	143	22.9	5.63	28.6	5	53	16
	18.1	175	90	8	5	9	141	23.0	5.63	28.2	5	53	17
175x125	23.3	169	125	8	5.5	12	129	29.7	7.81	23.5	5	70	20
175x175	32.8	171	174	9	6	12	129	41.7	9.67	21.5	5	94	21
	40.2	175	175	11	7.5	12	129	51.2	7.95	17.2	6	94	23
	40.4	175	175	11	7.5	13	127	51.4	7.95	16.9	6	94	24
200x100	17.8	198	99	7	4.5	8	168	22.7	7.07	37.3	4	57	15
	18.2	198	99	7	4.5	11	162	23.2	7.07	36.0	4	57	18
	20.9	200	100	8	5.5	8	168	26.7	6.25	30.5	5	57	16
	21.3	200	100	8	5.5	11	162	27.2	6.25	29.5	5	57	19
200x150	29.9	194	150	9	6	8	160	38.1	8.33	26.7	5	82	17
	30.6	194	150	9	6	13	150	39.0	8.33	25.0	5	82	22
	36.9	198	151	11	7	13	150	47.0	6.86	21.4	6	82	24
200x200	41.4	196	199	10	6.5	13	150	52.7	9.95	23.1	5	106	23
	49.9	200	200	12	8	13	150	63.5	8.33	18.8	6	106	25
	57.8	204	201	14	9	13	150	73.6	7.18	16.7	7	106	27
	65.7	208	202	16	10	13	150	83.7	6.31	15.0	7	106	29

Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
0.376	188	14.8	3.98	1.12	37.5	5.93	44.1	9.5	0.876	12.7	0.000	2.03
0.570	344	117.0	4.20	2.47	69.0	23.0	78.0	36.0	0.843	12.3	0.003	3.53
0.574	378	134	4.18	2.49	75.6	26.7	86.4	41.0	0.836	11.1	0.003	4.73
0.571	383	134	4.18	2.47	76.5	26.8	87.6	41.2	0.837	10.7	0.003	5.17
0.463	413	29.2	4.95	1.32	66.1	9.75	77.6	15.7	0.872	14.0	0.001	3.75
0.723	840	293	5.29	3.13	134	46.9	152	71.7	0.839	12.8	0.010	7.91
0.720	847	294	5.29	3.11	136	47.0	154	71.9	0.840	12.5	0.010	8.43
0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.003	2.81
0.670	1003	150	6.17	2.39	135	30.1	154	46.4	0.880	15.8	0.007	6.56
0.665	1021	151	6.17	2.37	138	30.1	157	46.7	0.881	15.0	0.007	7.37
0.872	1623	563	6.40	3.77	216	75.1	243	114	0.841	14.1	0.028	12.5
0.867	1641	563	6.39	3.75	219	75.1	246	115	0.842	13.7	0.028	13.5
0.877	2018	690	6.51	3.80	262	91.4	298	140	0.845	11.8	0.035	22.2
0.686	1205	97.5	7.26	2.06	138	21.7	156	33.6	0.885	21.8	0.007	4.31
0.685	1214	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.007	4.50
0.806	1529	261	7.18	2.97	181	41.8	202	64.3	0.879	19.1	0.017	6.76
1.01	2304	791	7.43	4.35	269	90.9	298	138	0.845	17.5	0.052	11.5
1.01	2884	984	7.50	4.38	330	112	369	171	0.844	14.7	0.066	20.4
1.01	2895	984	7.50	4.37	331	112	370	172	0.845	14.5	0.066	21.0
0.769	1543	113	8.25	2.24	156	22.9	175	35.5	0.883	28.4	0.010	3.30
0.764	1582	114	8.26	2.22	160	23.0	180	35.7	0.884	26.5	0.010	3.86
0.775	1806	134	8.23	2.24	181	26.7	205	41.6	0.878	24.9	0.012	5.08
0.770	1844	134	8.24	2.22	184	26.8	209	41.9	0.880	23.6	0.012	5.77
0.962	2625	507	8.30	3.65	271	67.6	301	103	0.876	21.2	0.043	9.32
0.954	2690	507	8.30	3.61	277	67.7	309	104	0.878	19.8	0.043	10.9
0.964	3331	633	8.42	3.67	336	83.8	377	128	0.879	16.9	0.055	18.5
1.15	3846	1315	8.54	5.00	392	132	433	201	0.847	18.2	0.114	17.6
1.16	4716	1602	8.62	5.02	472	160	525	244	0.846	15.5	0.142	29.8
1.17	5603	1897	8.73	5.08	549	189	617	287	0.847	13.6	0.171	45.8
1.18	6531	2201	8.83	5.13	628	218	710	332	0.848	12.2	0.203	66.7